Standing Waves

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Standing waves on a TL with different load

(i) $Z_R = \infty$:

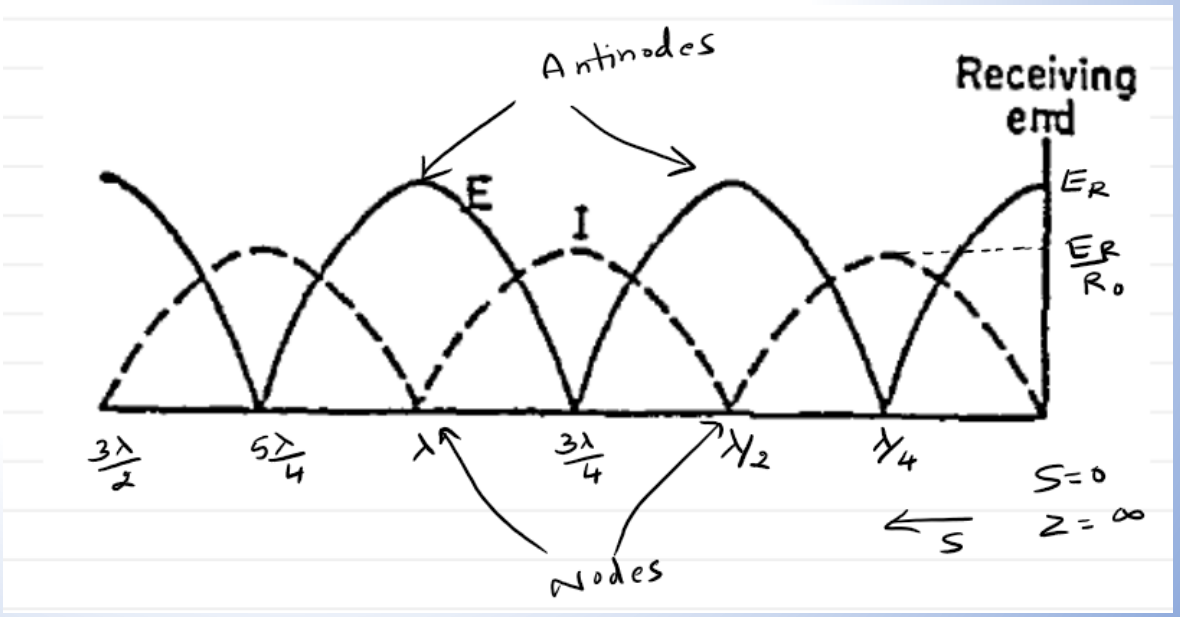
Given $E = E_R \cos \beta s + j I_R R_0 \sin \beta s$

$$I = I_R \cos \beta s + j \frac{E_R}{R_0} \sin \beta s$$

For $Z_R = \infty$, $I_R = 0$

$E = E_R \cos \beta s$

$I = j \frac{E_R}{R_0} \sin \beta s$



For λ/4

βs =(2π/λ)(λ/4)

= π/2

E = 0

$I = \frac{E_R}{R_0}$

- The current and voltages are in quadrature everywhere
- Thus no power is transmitted along the line
- Nodes and antinodes are used only for pure standing waves
- Current minima or voltage maxima occurs at 0, $\lambda/2$,λ… = n $\lambda/2$
- Current maxima or voltage minima occurs at $\lambda/4$,$3\lambda/4$… = (2n+1) $\lambda/4$

(ii) $Z_R = 0$:

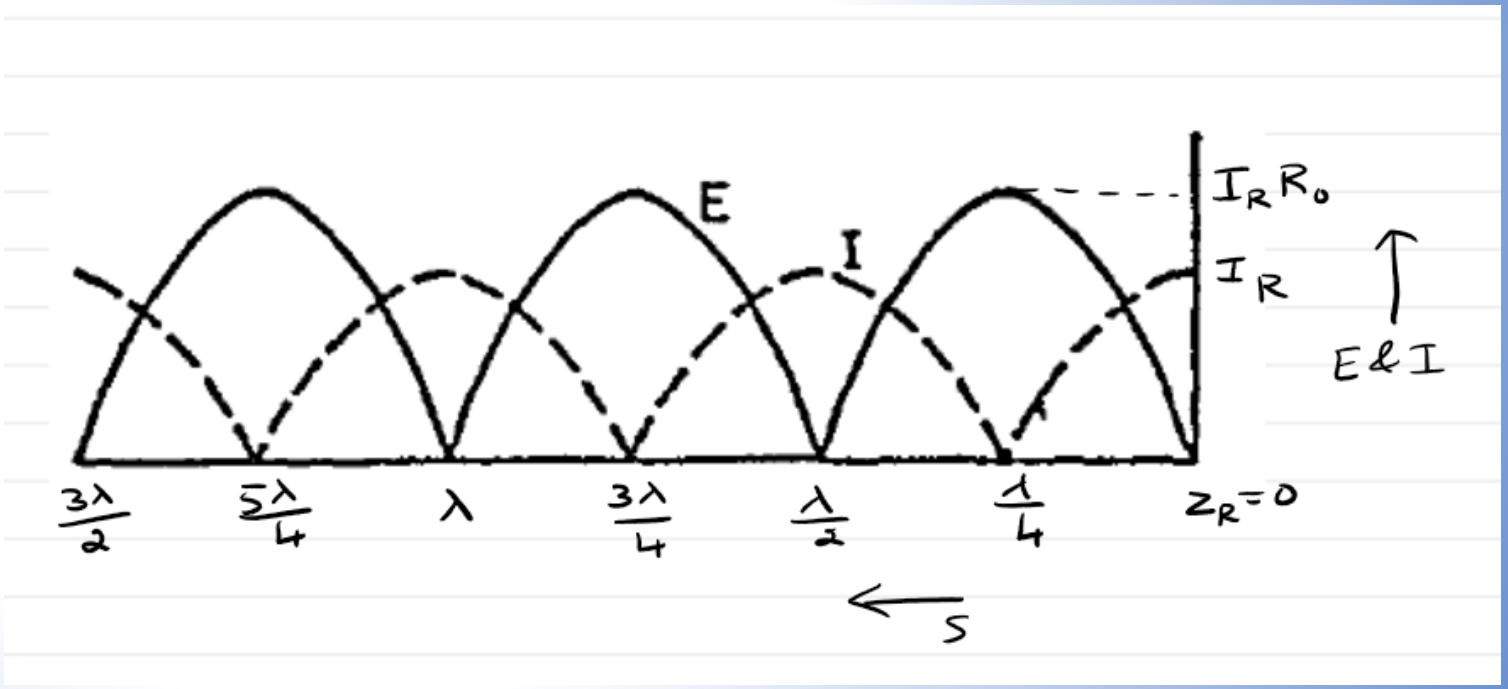
Given $E = E_R \cos \beta s + j I_R R_0 \sin \beta s$

$$I = I_R \cos \beta s + j \frac{E_R}{R_0} \sin \beta s$$



$E = j I_R R_0 \sin \beta s$

$I = I_R \cos \beta s$



- Current minima or voltage maxima occurs at $\lambda/4, 3\lambda/4 \ldots = (2n+1)\,\lambda/4$
- Current maxima or voltage minima occurs at $0, \lambda/2, \lambda \ldots = n\,\lambda/2$
- The above two cases are called pure standing waves, because all the signal is reflected

(iii) $Z_R = R_0$:

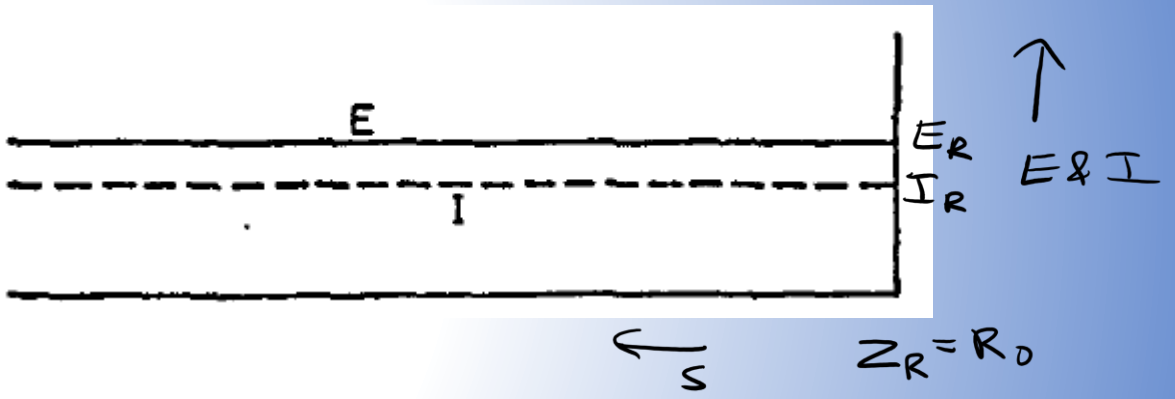
Given $E = E_R \cos \beta s + j I_R R_0 \sin \beta s$

$$I = I_R \cos \beta s + j \frac{E_R}{R_0} \sin \beta s$$



$E = E_R \cos \beta s + j E_R \sin \beta s = E_R e^{j\beta s}$

$I = I_R \cos \beta s + j I_R \sin \beta s = I_R e^{j\beta s}$



For $Z_R = R_0$, no reflected wave component

Maximum power will be delivered to the load

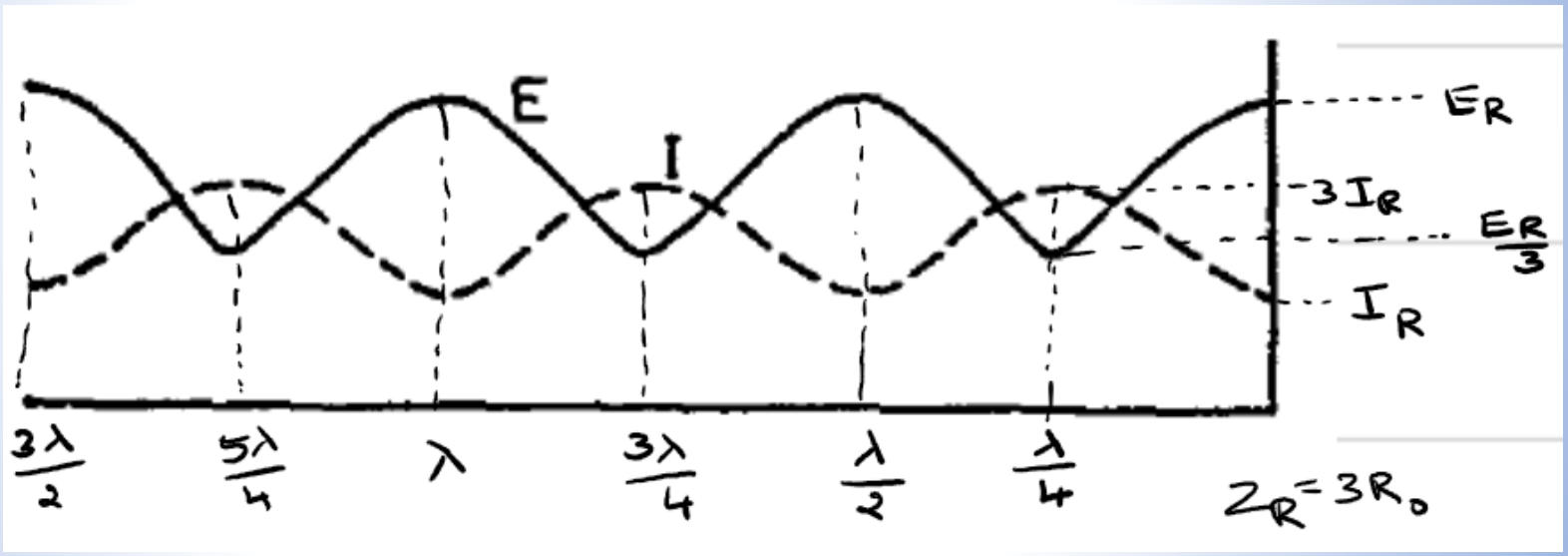
The lines are called “smooth” lines

(iv) $Z_R > R_0$:

Let $Z_R = 3R_0$

$E = E_R \cos \beta s + j\,(E_R/3)\sin \beta s = \frac{2}{3} E_R \cos\beta s + \frac{E_R}{3}(\cos\beta s + j \sin \beta s)$

$I = I_R \cos \beta s + j\, 3I_R \sin \beta s = I_R(\cos\beta s + j\sin\beta s) + j2I_R \sin\beta s$

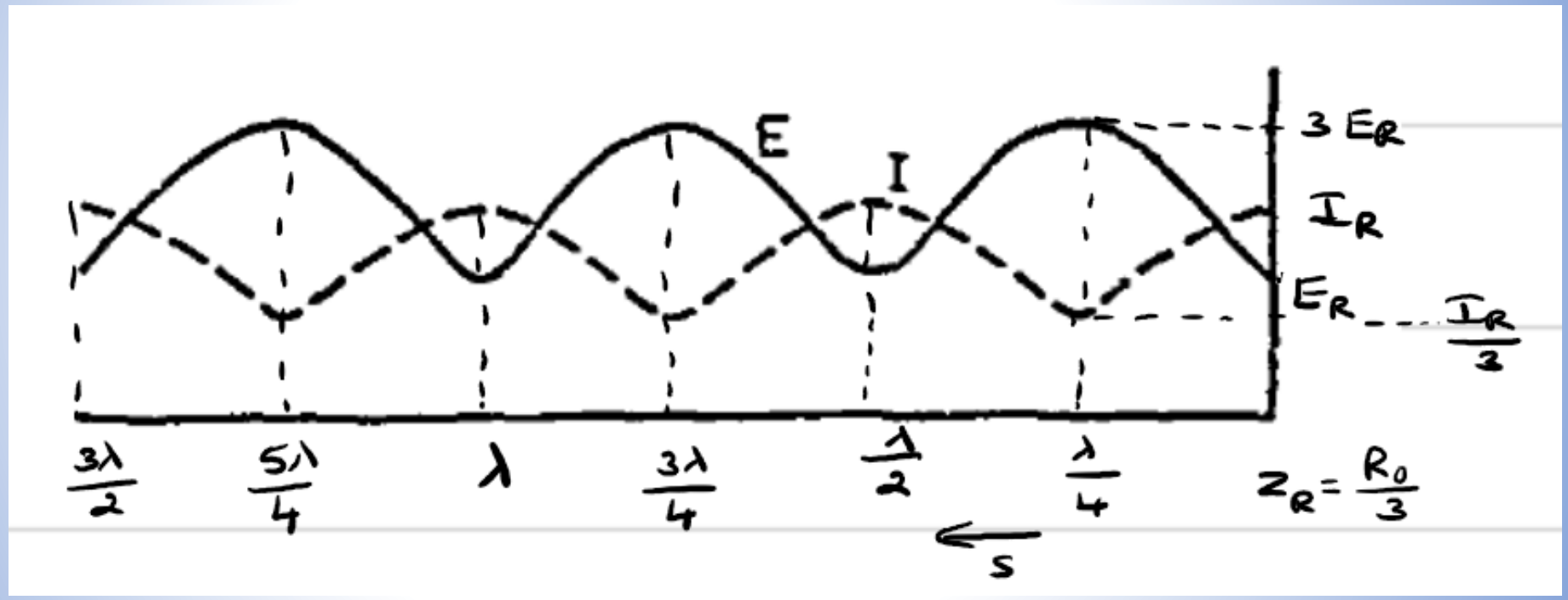


- Voltage and current minima occur at the voltage and current nodal points for an open circuited line
- It is not a pure standing wave (some reflection is present) and also some amount of power is delivered to the load

(v) $Z_R < R_0$:

Let $Z_R = R_0/3$

$$E = E_R e^{j\beta s} + j2E_R \sin\beta s \qquad I = \frac{2}{3} I_R \cos\beta s + \frac{I_R}{3} e^{j\beta s}$$



Voltage and current minima occur at the voltage and current nodal points for a short circuited line

Standing Wave Ratio (SWR)

- It is the ratio of maximum to minimum magnitudes of current or voltage on a line having standing waves

$S = \frac{|E_{max}|}{|E_{min}|} = \frac{|I_{max}|}{|I_{min}|}$ or

$VSWR = \frac{|E_{max}|}{|E_{min}|}$; $CSWR = \frac{|I_{max}|}{|I_{min}|}$

The maxima occur at points at which the incident & reflected waves are in phase and add directly

$$E = \frac{E_R}{2}\left(\frac{Z_R + R_0}{Z_R}\right)\left[e^{j\beta s} + K\, e^{-j\beta s}\right]$$

$$E_{max} = \frac{E_R}{2}\left(\frac{Z_R + R_0}{Z_R}\right)[1 + |K|]$$

Likewise, the voltage minima occur at points at which the incident & reflected waves are out of phase, thus

$$E_{min} = \frac{E_R}{2}\left(\frac{Z_R + R_0}{Z_R}\right)[1 - |K|]$$

- $S = \frac{|E_{max}|}{|E_{min}|} = \frac{1+|K|}{1-|K|}$ OR $|K| = \frac{s-1}{s+1}$
- This relation may be rearranged as

$$|K| = \frac{|E_{max}| - |E_{min}|}{|E_{max}| + |E_{min}|}$$

$|K| = |Er|/|Ei| = 1$ if $|Er| = |Ei|$

For $|K| = 0$, $S = 1$

For $|K| = 1$, $S = \infty$

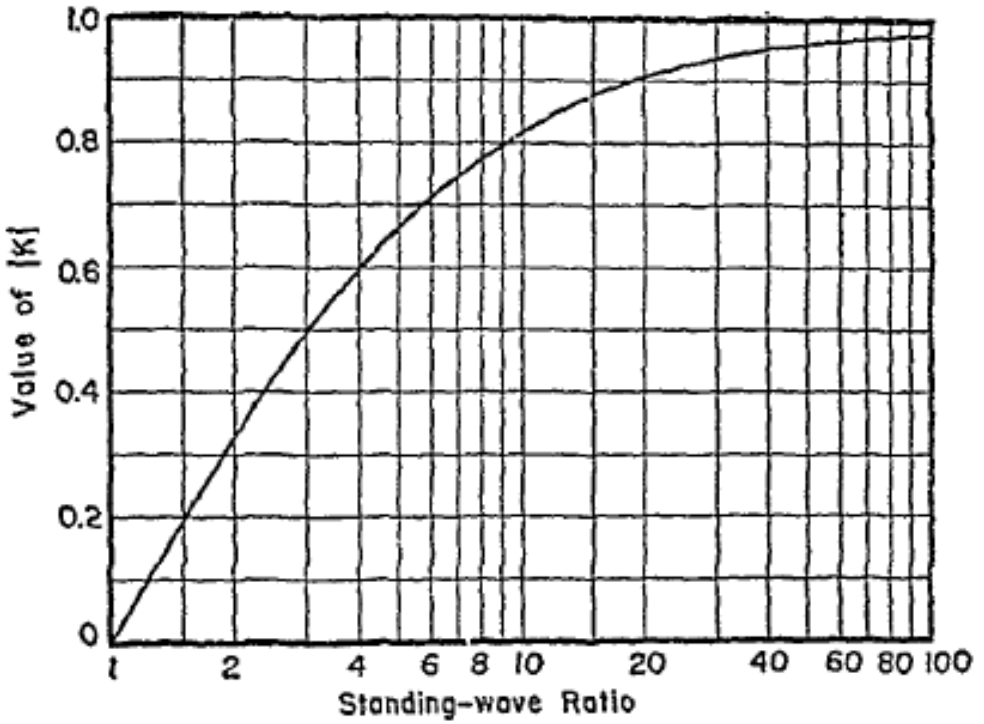


Fig. 7-8. Relation between the standing-wave ratio S and the magnitude of the reflection coefficient.

Problem : (3)

A transmission line has a characteristic impedance of $50+j0.01\ \Omega$ and is terminated in a load impedance of $73-j42.5\ \Omega$. Calculate (a) Reflection coefficient (b) Standing wave ratio.

(a) $$K = \frac{Z_R - Z_0}{Z_R + Z_0} = \frac{73 - j42.5 - (50 + j0.01)}{73 - j42.5 + (50 + j0.01)} = 0.377 \angle -42.7^\circ$$

(b) $$S = \frac{1+|K|}{1-|K|} = \frac{1+0.377}{1-0.377} = 2.21$$


Thank
you